# User Manual

## Deliverable Five

Group 19

# LEXXE

José Ribeiro (40745880)
Matthew Seery (44760442)
Oscar Gardner (44917651)
Thomas Pearce (43270484)
Tim Chard (44535686)

## Table of Contents

## Revision History

| Version | Date | Author | Description |
|---|---|---|---|
| 1.0 | 4 June, 2018 | Thomas Pearce | Initial document version. |

# Installation

## Packages

The software has a number of essential dependencies. These can be found in requirements.txt in the applications root directory.

The easiest way to ensure you have these dependencies is to first install anaconda3 and ensure that you are using this as your project interpreter.
From there, in a Linux environment, you can ensure the the remaining dependencies are installed by running the following command in the shell

```
conda install --yes --file requirements.txt
```

In a Windows environment, you may need to use the following command instead:

```
pip install -r ./requirements.txt
```

Should you have multiple Python versions running, it may be preferential to use a virtual environment. More info on how to do this can be found here:
https://conda.io/docs/user-guide/tasks/manage-environments.html

## Data

There are two ways the product can access data. Either through a JSON file, or a MongoDB. To switch between them, either specify “FILE” or “DB” in the “client” field of the configs (more on how to do this below). Should you choose to use MongoDB, ensure that you have an instance of MongoDB running, and that the connection parameters are properly set in your configuration.

You can find installation instructions for MongoDB here:

https://docs.mongodb.com/manual/administration/install-on-linux/
https://docs.mongodb.com/manual/tutorial/install-mongodb-on-windows/

Once you have chosen your data source, you must import the data. To do so, simply place your xml files (such as the ones your provided us) in a directory called “snippets” in the root directory of the project. From there, run `python ./main.py.`

And your data will be automatically imported into your data source of choice.

# Configuration

There are two ways to configure the solution. First is via command line arguments, and second is via the config.yml file in the projects root directory.

You can see the command line options for configuration by running

```
python ./main.py -h
```

As seen here:

```
Command Prompt

C:\Users\Tom>cd ./Documents/github/isys-358

C:\Users\Tom\Documents\Github\isys-358>python main.py -h
usage: main.py [-h] [-q {True,False}] [-m {BreakingNews,Keywords}] [-c CONFIG]
               [-bt BREAKING_THRESHOLD] [-dt DUPLICATE_THRESHOLD]
               [-kw KW [KW ...]] [-d] [-s {File,Mongo}] [-sd SNIPPETS_DIR]
               [-ld LOG_DIR] [--db_host DB_HOST] [--db_port DB_PORT]
               [--db_name DB_NAME]

Breaking News Analysis

optional arguments:
  -h, --help            show this help message and exit
  -q {True,False}, --quick {True,False}
                        Skips cpu intensive operations to get a quick but
                        approximate result
  -m {BreakingNews,Keywords}, --mode {BreakingNews,Keywords}
                        The mode to run on the articles.
  -c CONFIG, --config CONFIG
                        Path to yaml configuration file. Default: 'config.yml'
  -bt BREAKING_THRESHOLD, --breaking_threshold BREAKING_THRESHOLD
                        The threshold used to determine if an article is
                        breaking news.
  -dt DUPLICATE_THRESHOLD, --duplicate_threshold DUPLICATE_THRESHOLD
                        The threshold used to determine if two articles are
                        duplicates.
  -kw KW [KW ...], --keywords KW [KW ...]
                        Keywords to run process
  -d, --debug           Turns on debug logging
  -s {File,Mongo}, --storage {File,Mongo}
                        The method used to store temprory data.
  -sd SNIPPETS_DIR, --snippets_dir SNIPPETS_DIR
                        Path of directory which contains raw article snippets
  -ld LOG_DIR, --log_dir LOG_DIR
                        Path of directory where logs will be stored.
  --db_host DB_HOST     IP Address of the MongoDB host (ignored unless Mongo
                        is passed as -s argument)
  --db_port DB_PORT     Port used to communicate with MongoDB (ignored unless
                        Mongo is passed as -s argument)
  --db_name DB_NAME     Name of the MongoDB collection (ignored unless Mongo
                        is passed as -s argument)

C:\Users\Tom\Documents\Github\isys-358>
```

The command line is preferential in many cases, especially should you want to change the configuration for a single run, however if you want to set up a default configuration that you expect to use in the future, modification of the config.yml file is suggested. Should you not know what certain parameters are for, simply consult the output of the help command as shown above.

# Running

To run the solution, navigate to the project's root directory and simply use the command

```
python ./main.py
```

If you want to use a certain configuration, see the configuration section above.
You can also pass the output directly to the evaluation program with the following command

```
python ./main.py | python ./evaluator.py
```

# Tests

There are 3 test files we have for testing the functionality of the solution. These are data.py processors.py and utils.py and they can all be found in the test folder in the root directory. These tests are unit tests that cover key functions in order to ensure correct behaviour.

Should you wish to run these tests, you can do so by running

```
python -m unittest
```

# Troubleshooting

**Missing packages/wrong Python version**
If you are seeing issues about missing packages, or having the wrong python version, then it is a good idea to test you are using the correct interpreter.
You can do this by running (the V is case sensitive)

```
python -V
```

If you followed the advice to use Anaconda as the project interpreter, you should see the output:

```
Python 3.6.4 :: Anaconda, Inc.
```

**Cannot open file ‘main.py’**

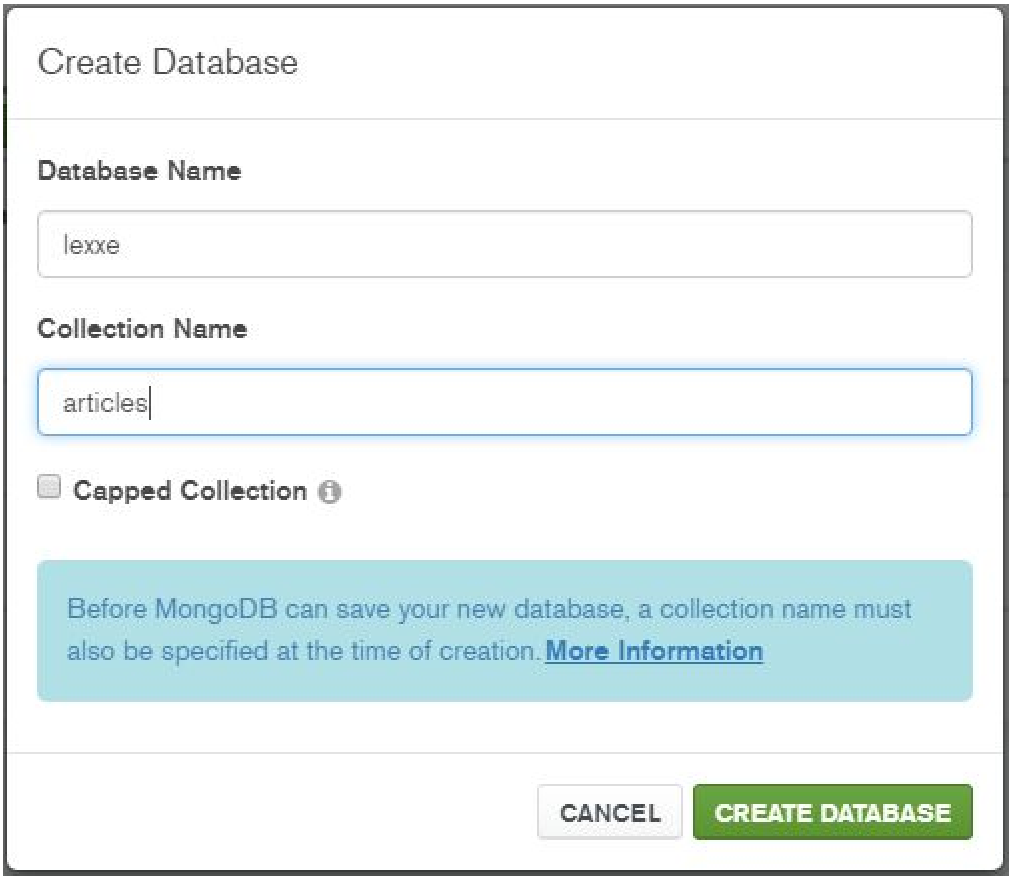
If you see output similar to:

```
python: can't open file 'main.py': [Errno 2] No such file or directory
```

Please ensure you have navigated to the root directory of the project before running

```
python ./main.py
```

**Collection “Lexxe” is missing or unavailable**
Under certain configurations, Pymongo may not automatically create the database and collection. In such a scenario, simply use compass or the command line to create a “Lexxe” database, and “articles” collection like so:


Create Database
Database Name
lexxe
Collection Name
articles
Capped Collection
Before MongoDB can save your new database, a collection name must also be specified at the time of creation. More Information
CANCEL
CREATE DATABASE